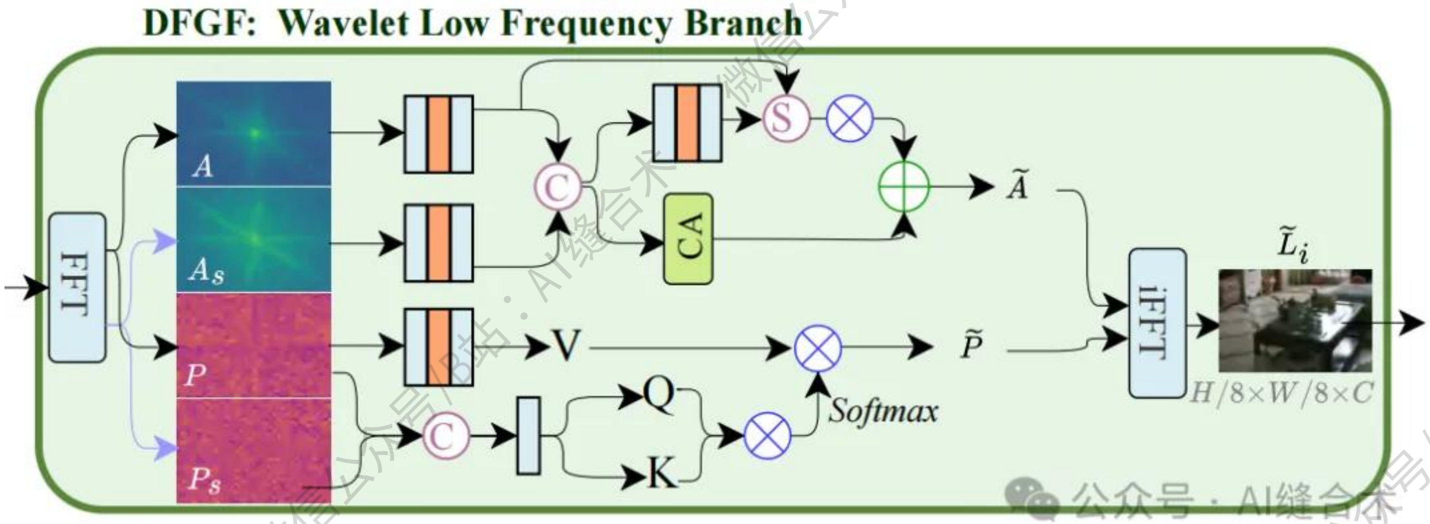
核心速览



论文题目：SPJFNet: Self-Mining Prior-Guided Joint Frequency Enhancement for Ultra-Efficient Dark Image Restoration

中文题目：SPJFNet：自挖掘先验引导联合频率增强用于超高效暗图像恢复

论文链接：https://arxiv.org/pdf/2508.04041

所属单位：吉林大学

核心速览：

本文针对暗图像恢复中存在的外部先验依赖、多阶段冗余操作及频域无差别处理导致的效率瓶颈，提出高效自挖掘先验引导联合频率增强网络（SPJFNet）。该网络通过自挖掘引导模块（SMGM）生成内生先验以消除外部

依赖，结合双频引导框架（DFGF）实现小波域高频细节增强与傅里叶域低频光照恢复的协同，在提升恢复质量的同时显著降低计算复杂度，实验验证其性能优于现有方法。

论文内容详细解读：

https://mp.weixin.qq.com/s/OJnnvNKIYVMdiJD5YLFulQ

```
        (2): Conv2d(4, 32, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (3): Sigmoid()
      )
    )
    (concat_layers): ModuleList(
      (0-2): 3 x Sequential(
        (0): Conv2d(32, 16, kernel_size=(1, 1), stride=(1, 1))
        (1): SEBlock(
          (fc): Sequential(
            (0): AdaptiveAvgPool2d(output_size=1)
            (1): Conv2d(16, 4, kernel_size=(1, 1), stride=(1, 1))
            (2): ReLU(inplace=True)
            (3): Conv2d(4, 16, kernel_size=(1, 1), stride=(1, 1))
            (4): Sigmoid()
          )
        )
      )
    )
    (recon_trunk): Sequential(
      (0): ResidualBlock_noBN(
        (conv1): Conv2d(16, 16, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
        (conv2): Conv2d(16, 16, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
      )
    )
    (upconv_last): Conv2d(16, 3, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (lrelu): LeakyReLU(negative_slope=0.1, inplace=True)
  )
  (conv_first_fr): Conv2d(3, 16, kernel_size=(1, 1), stride=(1, 1))
  (conv_first_map): Conv2d(3, 16, kernel_size=(1, 1), stride=(1, 1))
)
input_tensor_shape_x1  : torch.Size([1, 3, 64, 64])
input_tensor_shape_x2  : torch.Size([1, 3, 64, 64])
output_tensor_shape : torch.Size([1, 3, 64, 64])

哔哩哔哩/微信公众号：AI缝合术，独家整理！
```